Blocaïne

Installation & Evaluation Procedure

(The open source solution for automation)

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1 General information

To evaluate **Blocaïne**, you only need a single PC running two Python interpreters, one for the Blocaïne **Interpreter** and one for the **Block Editor.**

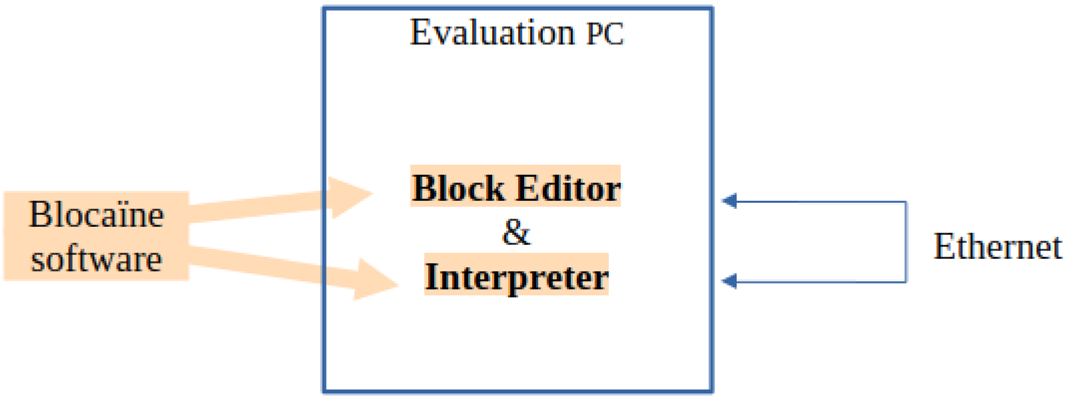


Figure 1 : Evaluation Architecture

2 Prerequisites

You need a computer with at least 200 Mo of free disk and Python (version 3.6 or higher) programming language installed.

3 Installation of Blocaïne

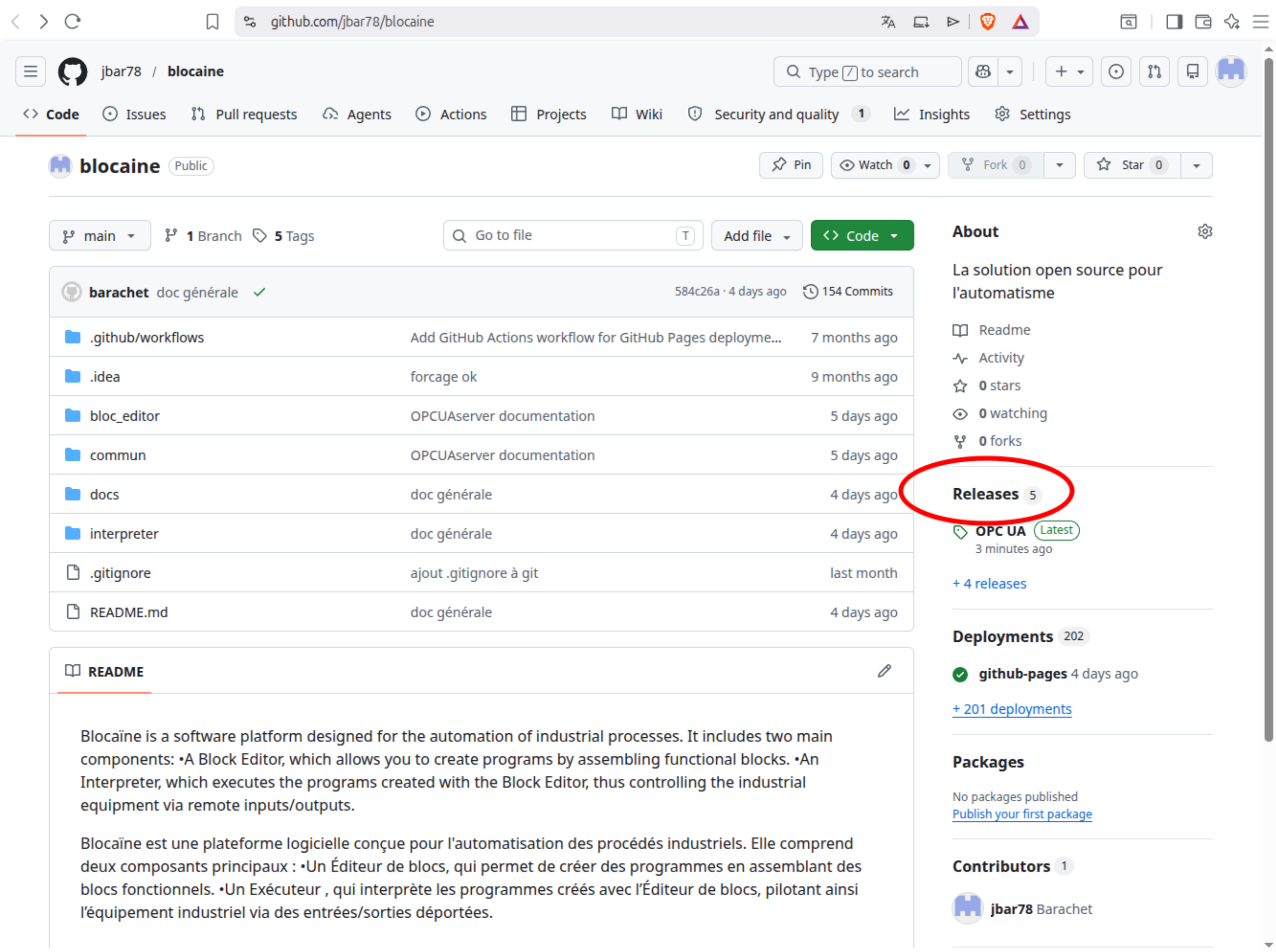
Strictly speaking, there is no real installation process. You simply need to download a ZIP file from GitHub and unzip it. Follow the steps below :

3.1 Downloading Blocaïne

Using your browser, go to:

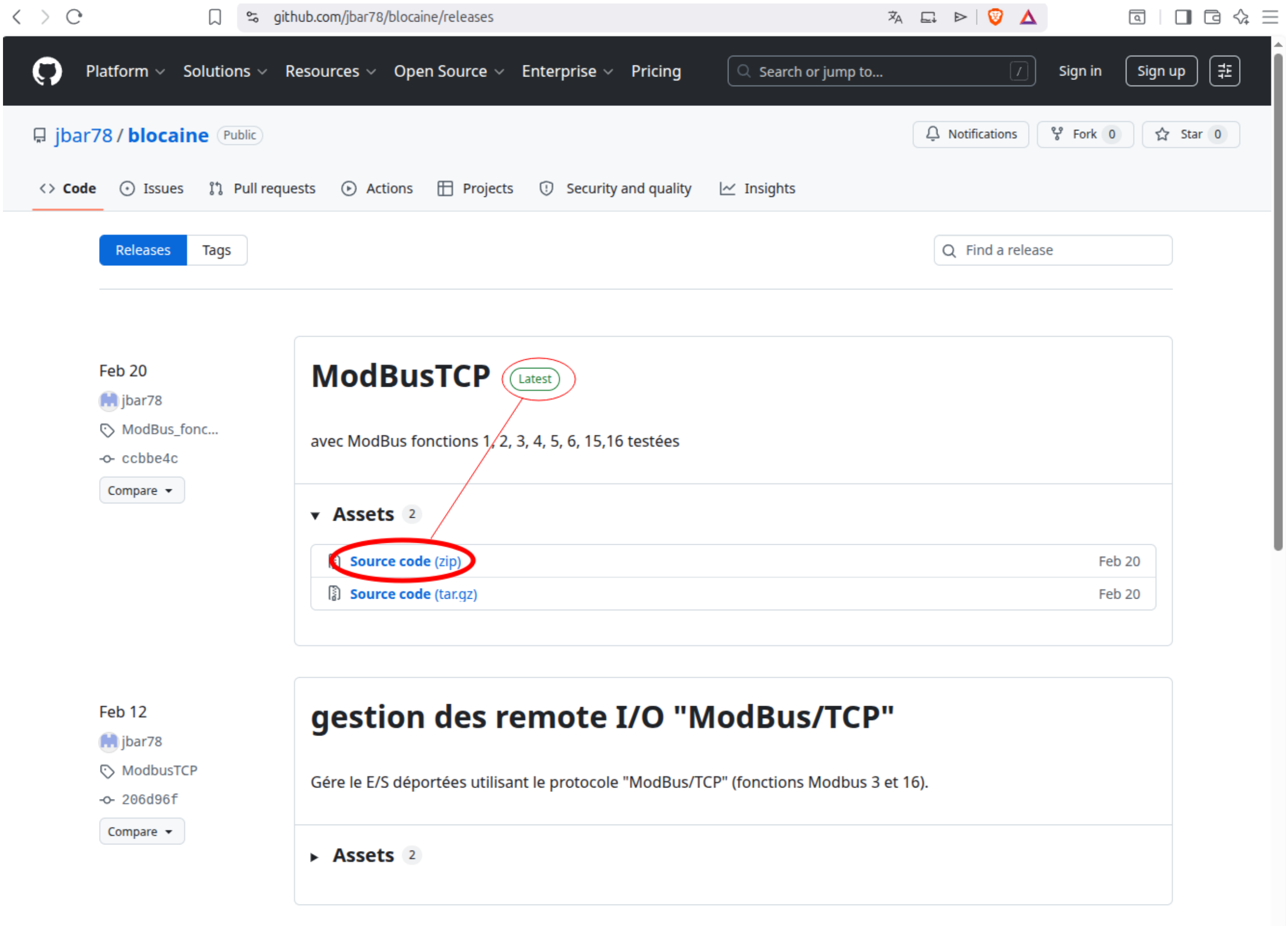
https://github.com/jbar78/blocaine.git

You will arrive on the following page :



Click on « Releases ».

A page like this will appear. :



Click « Source code (zip) » of "latest" version, select here you want to save ZIP file and once downloaded, **unzip** it. you should then have a new directory named «blocaine-*x.y* ». (*x.y* is the release version)_

3.2 Installing Python Libraries

You don't need to do anything, since the version of Blocaïne you have just downloaded only uses Python3 standard library, because the ModBus and OPC features have been disabled to make installation easier.

Blocaïne is therefore ready to use. You can go to "Launching the **Interpreter**" chapter and then to "Launching the **block editor"** chapter.

3.2.1 Using ModBus I/O

If you want to use "ModBus" hard I/O, you must enable this feature by editing the file `PARAM_CONFIG.py` located in the `Target` directory, changing the value of the `PARAM_CONFIG_MODULE_MODBUS` parameter from `False` to `True`, then copy this file into the root directory `blocaine-`*x.y* . You must also install the following library (module): `pymodbus` as follows:

Under Windows, type in the console:

```
pip install pymodbus
```

Under Ubuntu, type in the console:

```
sudo apt install python3-pymodbus
```

Under other Linux distributions, type:

```
python3 -m pip install pymodbus
```

3.2.2 Using the OPC Protocol

If you want the Blocaïne executor to include an OPC UA server so that it can connect to your HMI, you must enable this feature by editing the file `PARAM_CONFIG.py` located in the `Target` directory, changing the value of the `PARAM_CONFIG_MODULE_OPCUA` parameter from `False` to `True`, then copy this file into the root directory `blocaine-`*x.y*. You must also install the following library (module): `opcua` as follows:

Under Windows, type in the console:

```
pip install OPCUA
```

Under Ubuntu, type in the console:

```
sudo apt install python3-OPCUA
```

Under other Linux distributions, type:

```
python3 -m pip install opcua
```

4 Network Configuration

You don’t need to do anything, since the Target’s IP address of HTTP and TCP/IP servers of **interpreter** are by default set to: 127.0.0.1

Note: the target address is defined by the parameter PARAM_TCP_TARGET_IP = "127.0.0.1" contained in the `PARAM_NETWORK.py` files located in the “commun” directory.

5 Launching the **Interpreter**

5.1 Under Windows

Open the folder "blocaine-*x.y*\interpreter", double-click "main.py." If the firewall detects new connections, click Allow.

5.2 Under Linux

Open the folder "blocaine-*x.y*/interpreter" in the file explorer, right-click on "launcher_interpreter.sh", then select "Run as a program". Prefer option 2 if you know the "sudo" password; otherwise, use option 1. The Python3 interpreter will start, and a terminal window will appear.

5.3 Under MacOS

Run "main.py" inside the "blocaine-*x.y*/interpreter" directory using the Python3 interpreter.

Once launched, the **Interpreter** is operational. You can now open your web browser and go to http://127.0.0.1:8080 (port 8080) to navigate through the **Interpreter**'s menu

6 Launching the **Block Editor**

6.1 Under Windows

Open “blocaine-*x.y*\bloc_editor”, then double-click “main.py.”

6.2 Under Linux

Open the “blocaine-*x.y*/bloc_editor” directory in your file explorer, right-click on “launch_bloc_editor.sh," then select “Run as a program.”

6.3 Under MacOS

Run “main.py” in the “blocaine-*x.y*/bloc_editor” project directory using Python3

The Python interpreter will start, and both a terminal and the **Block Editor**’s graphical interface will appear. The **Block Editor** automatically connects to the Blocaïne **interpreter**.

You can now create your first **user** blocks, upload them, and execute them in the **interpreter.**

7 Verify that Blocaïne is working properly

7.1 Checking the connection

To check whether the **interpreter** is properly connected to the **Block Editor**, look at the status bar located at the bottom of the **Block Editor**'s graphical interface. It should display the message "Target: Connected."

7.2 Editing the user block <loop>

Using the **Block Editor**, edit the user block <loop> via the "Bloc" menu. Then click "Open," select the file "loop.bloc," and click "Open." The block should appear in the graphical area.

7.3 Compilation, download & execution

To compile the user block <loop>, download it to the Executor, and run it, use the "Target" menu. Then click "Check, Transfer & Execute <loop>" or click the ⚡ icon.
After compilation, a popup will ask you to confirm whether you want to download and run the <loop> block. Click "Yes."
You can verify that the <loop> block has started correctly by using the web server (http://127.0.0.1:8080).

7.4 Debugging

To view the current values of the variables in the <loop> block, enable "Monitoring" mode via the "Target" menu. Then click "Monitoring (Start/Stop) [Space]," or press the "Space" key, or click the 👁️ icon.

8 Uninstalling **Blocaïne**

Since installation only involves downloading and unzipping a ZIP file to obtain the “blocaine-*x.y*” directory, uninstalling Blocaine simply consists of deleting the ZIP file and the “blocaine-*x.y*” folder.